

Introduction

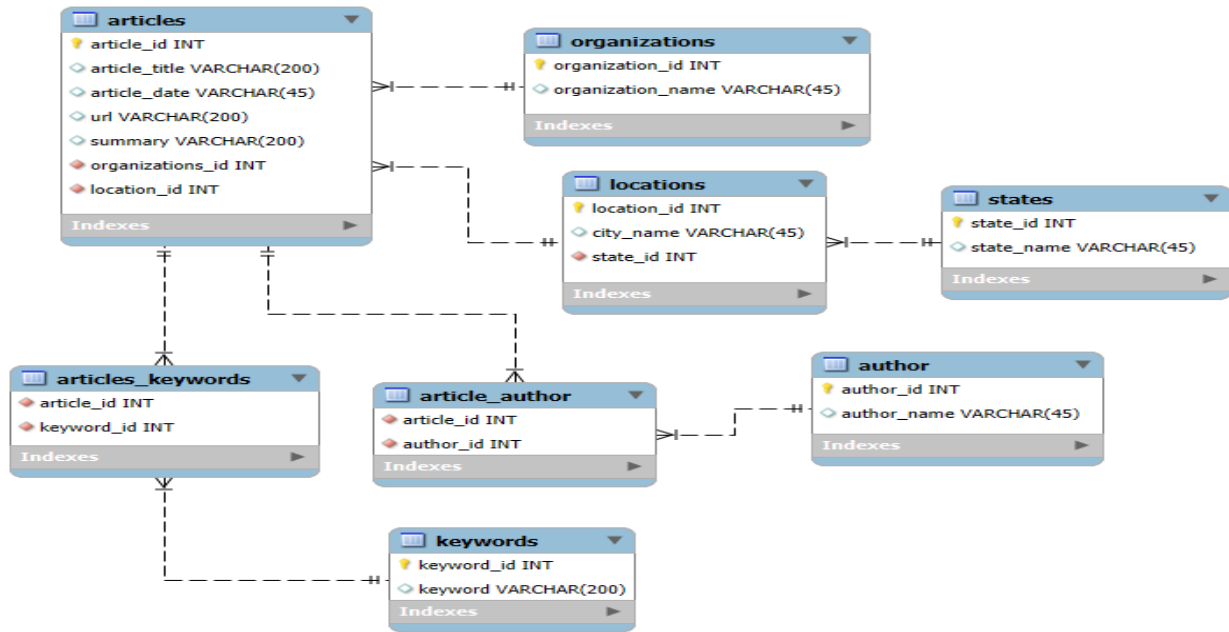
Hate crimes remain a significant societal issue, often correlated with political events and cultural shifts. Following the 2016 U.S. Presidential Election, reports of hate crimes notably increased, prompting the need for structured data analysis. This project focuses on media coverage of hate crimes from March to August 2017. By designing and implementing a well-structured relational database, we aim to uncover geographic and temporal trends in hate crime reporting. This database can serve as a resource for researchers, policymakers, advocates, and the public, making it easier to track patterns, inform policy decisions, and raise awareness.

Database Description

Our database analyzes how media outlets cover hate crimes, emphasizing geographic patterns and temporal trends. The database schema consists of eight tables. The main “articles” table stores key information, including article ID, title, date, URL, summary, organization ID, and location ID. We use junction tables, such as “article_keywords” and “article_author,” to manage many-to-many relationships between articles and keywords, as well as articles and authors. The schema also includes one-to-many relationships, “articles” to “organizations” and “articles” to “locations”, as well as “locations” to “states,” ensuring a fully normalized structure (3NF) that limits redundancy and supports efficient queries.

Logical Design

Our ERD (Entity-Relationship Diagram) matches the physical database structure, illustrating all entities, their attributes, and the relationships between tables. The ERD shows the “articles” table at the center, connected to auxiliary tables such as “organizations,” “locations,” “states,” “keywords,” and “authors.” Many-to-many relationships are supported by dedicated junction tables. Primary and foreign keys ensure referential integrity. The final ERD was refined after receiving TA feedback to ensure clarity in the relationship lines and correct cardinalities.



Note: The ERD model was changed based on the feedback from multiple TAs with different opinions on how the relationship lines should appear, affecting the primary and foreign keys.

Physical Database

Our database is designed to support researchers, policymakers, and advocates in analyzing hate crime trends by structuring the data to prioritize usability and flexibility. The scope of the "Documenting Hate" dataset informed the physical design of our database. For example, we limited our data to articles between March and August 2017, selecting entries with sufficient details such as authorship, summaries, and keywords to ensure meaningful analysis. We organized the schema into eight normalized tables to meet user needs effectively. Researchers exploring geographic trends require a clear connection between articles and locations, so we created a locations table linked to states and the main articles table through foreign keys. To address questions about patterns in reporting, such as frequently used terms or recurring contributors, we designed junction tables (article_keywords and article_authors) to manage many-to-many relationships between articles, keywords, and authors.

Sample Data

article_id	article_title	article_date	url	summary	organization...	location_id
1	New Kentucky Law: Attacking A Police Officer Is...	2017-03-24 14:56:00	http://www.westernjournalism.com/new-kentuck...	by Fred Maxwell March 24, 2017 at 2:55pm K...	1	1
2	Future of hate crime prosecutions questioned	2017-03-24 19:26:00	http://www.houmatoday.com/news/20170326/fut...	By Ann E. Marimow and Rachel Weiner The Wa...	2	2
3	Report Hate Speech, Law Enforcement Officials...	2017-03-26 16:37:00	http://kcur.org/post/report-hate-speech-law-enfo...	By Laura Ziegler 6 hours ago Hate speech...	3	3
4	Newsday letters to the editor Monday, March 27...	2017-03-27 07:45:00	http://www.newsday.com/opinion/letters/letters-e...	By Newsday Readers A pedestrian walks acros...	4	4
5	City Of Tucson Expands Hate-Crime Resolution	2017-03-27 15:52:00	http://kjzz.org/content/451800/city-tucson-expan...	By Mark Brodie Last week, the city of Tucson p...	5	5
6	Killings highlight marginalization of transgender...	2017-03-29 09:52:00	http://www.humanosphere.org/human-rights/201...	By Lisa Nikolau on 29 March 2017 0 Crosse...	6	6
7	GoFundMe Page Set Up In Support of the Irani...	2017-03-31 10:26:00	http://www.eweek.com/news/2017/03/31/gofund...	On Tuesday, Hasel Afshar came home from a s...	7	7

The data in our database was extracted by using pandas, which we operated within Jupyter Lab to process the original CSV file. First importing the Documenting Hate.csv file into Pandas, ensuring the correct encoding to handle special characters. Next, we cleaned the data by removing any non-ASCII characters and dropping rows with missing values to ensure completeness. To standardize the information, we converted full state names to their two-letter abbreviations and created separate tables for states, locations, organizations, authors, and keywords. We extracted author names from the article summaries using regular expressions and organized them into an author's table. Additionally, we processed the keywords by cleaning and standardizing them, then linking them to their respective articles through a join table. Finally, we structured the main articles table by linking each article to its corresponding organization and location using unique identifiers. This streamlined process ensured our database was well-organized and ready for effective analysis of hate-related incidents. The image above is a sample of data from our articles table.

Views and Queries

Our database includes five views that provide various insights into the hate crime articles data:

1. articles_with_authors_and_orgs
 - a. Purpose: Combines information from articles, authors, and organizations.
 - b. Usage: Provides a comprehensive overview of each article with its authors and publishing organization.
 - c. SQL Features: Joins multiple tables, uses GROUP_CONCAT for author aggregation.
2. Articles_with_keywords
 - a. Purpose: Shows articles published from June-August 2017 with their associated keywords.
 - b. Usage: Allows analysis of trending topics during a specific time period. This period was chosen as it represents a peak in hate crime reporting following significant events.
 - c. SQL Features: Joins multiple tables, uses date filtering, GROUP_CONCAT for keyword aggregation.
3. articles_per_state

- a. Purpose: Counts articles for each state.
 - b. Usage: Identifies which states have the most coverage or incidents related to hate crimes.
 - c. SQL Features: Joins multiple tables, uses COUNT for aggregation, ORDER BY for ranking.
- 4. Top_authors
 - a. Purpose: Identifies authors who have written the most articles on hate crimes.
 - b. Usage: Highlights key contributors or experts in hate crime reporting. Authors are ranked by the number of articles written, with a minimum threshold of 5 articles.
 - c. SQL Features: Uses subquery, COUNT for aggregation, HAVING clause for filtering.
- 5. Hate_crime_articles
 - a. Purpose: Focuses on articles specifically related to hate crimes.
 - b. Usage: Tracks and analyzes coverage of hate crime incidents over time.
 - c. SQL Features: Uses subquery, keyword matching with LIKE, ORDER BY for chronological sorting.

We also created two additional non-view related queries that provide interesting insights:

- 6. Monthly Trend of Hate Crime Reporting
 - a. Purpose: Visualizes the frequency of hate crime reporting over time.
 - b. Identifies patterns or spikes in hate crime coverage, which could correlate with real-world events.
- 7. Geographic Distribution of Hate Crime Reporting
 - a. Purpose: Maps the distribution of hate crime articles across different states.
 - b. Usage: Highlights regional variations in hate crime reporting, potentially indicating areas of higher incidence or media focus.

View Name	Req. A	Req. B	Req. C	Req. D	Req. E
articles_with_authors_and_orgs	X			X	

articles_with_keywords	X	X			
articles_per_state	X		X		
organization_article_stats	X	X	X		X
hate_crime_articles	X	X			

Changes from Original Design

Our original design included a timeline covering every month from February to August 2017. However, we realized that the main "Documenting Hate.csv" dataset was incomplete, with many summaries and keywords missing. To address this, we shortened our timeline to focus on data from March to August and ensured consistency by cleaning the data to meet proper encoding standards (UTF-8) and formatting styles, as detailed in the sample data section. Additionally, we refined our schema significantly. The original design contained 14 tables, many of which were redundant or unnecessary. Through normalization, we streamlined the design to 8 tables, ensuring all relationships adhered to the third normal form (3NF). Key changes included consolidating cities and states into a single locations table and introducing junction tables (article_keywords and article_authors) to handle many-to-many relationships. These adjustments created a more efficient and scalable database while retaining the ability to support detailed trend analysis.

Database Ethics Considerations

Our database prioritizes the privacy and confidentiality of individuals and organizations. It includes details like locations, dates, and keywords but deliberately excludes personally

identifiable information about victims and perpetrators to protect privacy and minimize risks of re-identification or harm. This approach ensures the database remains a tool for research, advocacy, and journaling without perpetuating discrimination or bias. To address fair use and copyright concerns, we used data exclusively from public reports, reputable news outlets, and authorized nonprofit organizations, ensuring compliance with copyright laws (Documenting Hate csv). Proprietary or closed-source information was excluded to avoid legal liabilities. Access to the database is restricted to authorized users for legitimate research and analysis, with ethical guidelines ensuring unbiased, research-focused use. Additionally, user anonymity is safeguarded by excluding sensitive data like victim names and exact addresses. In cases where sensitive information might be exposed, explicit consent will be required. By focusing on privacy, compliance, and ethical data use, our project supports efforts to reduce hate crimes responsibly and effectively.

Lessons Learned

The greatest lesson we learned during this project was the importance of time management and collaboration. Balancing schoolwork, jobs, and other priorities made it challenging to find consistent times to meet and work on the project. Using zoom and group chats helped us delegate tasks and allowed us to tackle different portions of the workload efficiently. One of the key technical lessons was learning how to import and organize data effectively. By using pandas in Jupyter Lab, we cleaned the dataset to remove non-ASCII characters, applied proper UTF-8 encoding, and ensured consistency by standardizing fields like state names. Extracting and organizing sample data taught us how to break down large datasets into meaningful subsets. For example, we filtered articles with authors to improve the quality of analysis and linked important groups of information—such as keywords and authors—to articles through intermediary tables. We also struggled with normalization, in the beginning we struggled with entities in wrong tables. Working as a group and TA, we reorganized our schema to the 3NF. Additionally, the process of querying data was invaluable in understanding what information we needed. This helped us grasp how to structure and analyze the data while working together.

Potential Future Work

For more improvement to the database, we could collectively ensure that the minor details are refined. Additionally, our data scope could be further expanded beyond the years that

we researched to provide a long-term, more specific perspective on hate crime trends. To target more specific crime details, an “incident” table named “victim demographics” for example, would allow for a deeper understanding of the crimes with better context rather than the articles simply reporting them. An additional table, named “media type” could categorize the types of organizations that report how different types of media convey hate crimes from differing perspectives.

Looking further, the database could be adapted into a web application for broader accessibility. The app could allow users to filter and visualize hate crime data through interactive maps and graphs. Features like notifications for updates on recent hate crime reports or the ability to share insights with others could enhance its usability for researchers, policymakers, and community advocates.

Citations

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